
SOFTWARE REQUIREMENT

Finding the voice of the user

(chapter 06)

Objectives

- Student should understand that the success in software requirements, and hence in software development, depends on getting the voice of the user close to the ear of the developer.
- Student should enhance step by step to find the voice of the user
- Student could classify users and try to understand what type of information the users from each department might supply based on their role in the organization and their department's perspective on the projects
- Enhance the differences in user representation on agile projects

Contents

- User classes
- User personas
- Connecting with user representatives
- The product champion
- User representation on agile projects
- Resolving conflicting requirements

Finding the voice of the user

- Success in software requirements, and hence in software development, depends on getting the voice of the user close to the ear of the developer
- Steps to find the voice of the user
 - Identify the different classes of users for your product
 - Select and work with individuals who represent each user class and other stakeholder groups
 - Agree on who the requirements decision makers are for your project.
- Customer involvement is the best way to avoid the expectation gap a mismatch between the product that customers expect to receive and what developers build
- The features that users present as their “wants” don’t necessarily equate to the functionality they need to perform their tasks with the new product

User Classes

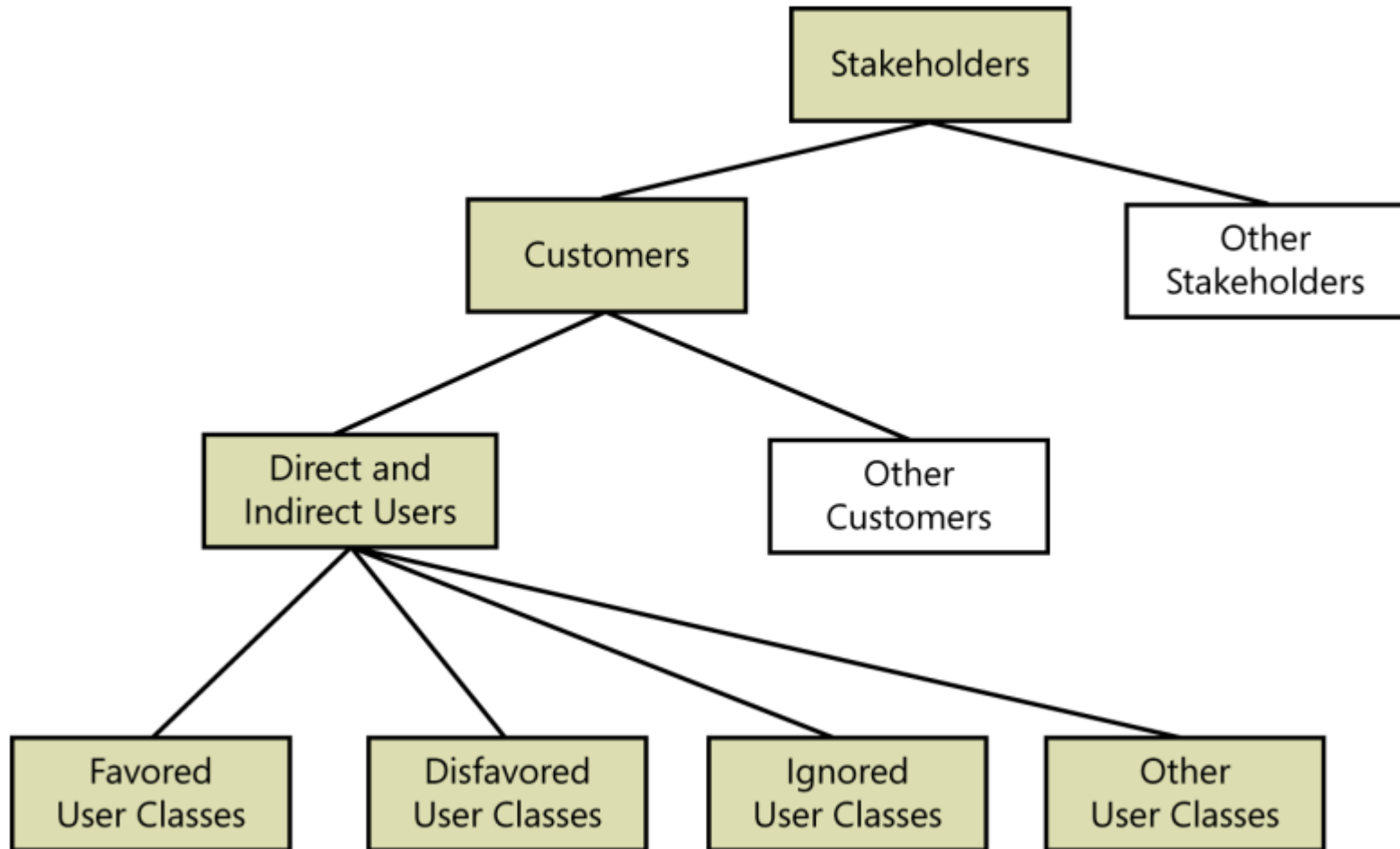
Classifying Users

You can group users into a number of distinct *user classes* based on these sorts of differences:

- Their access privilege or security levels (such as ordinary user, guest user, administrator)
- The tasks they perform during their business operations
- The features they use
- The frequency with which they use the product
- Their application domain experience and computer systems expertise
- The platforms they will be using (desktop PCs, laptop PCs, tablets, smartphones, specialized devices)
- Their native language
- Whether they will interact with the system directly or indirectly

User Classes

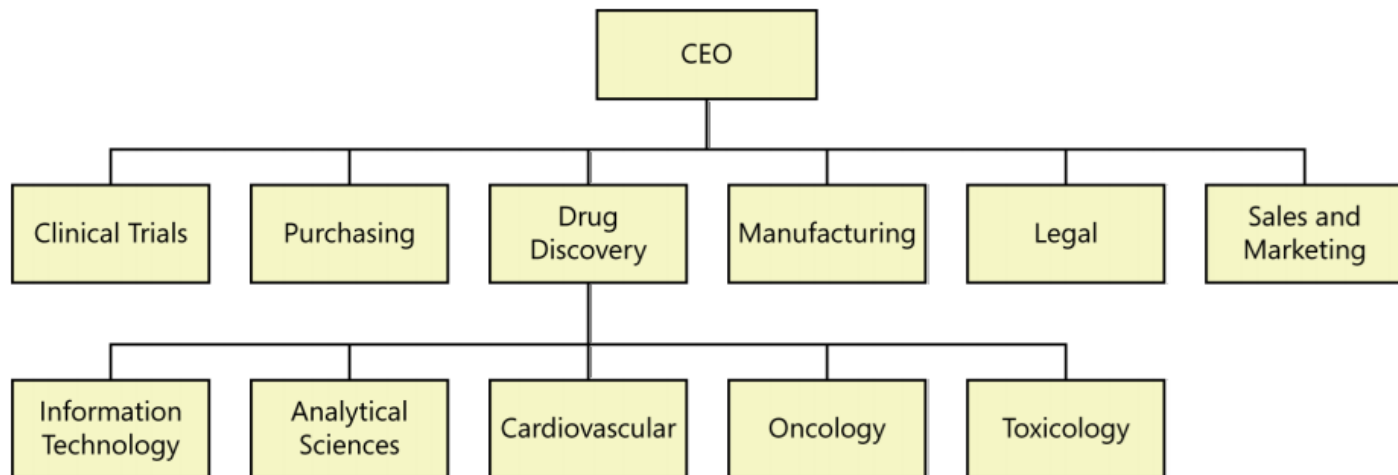
Classifying Users



User Classes

Identifying user classes for your project

- A useful technique for this is a collaboration pattern
 - Ask the for the list of people who he expects to use system
 - Brainstorm as many user classes as you can think of
 - Look for groups with similar needs that you can either combine or treat as a major user class with several subclasses
- The external entities shown outside your system on a context diagram are candidates for user classes.
- A corporate organization chart can also help you discover potential users and other stakeholders



User Personas

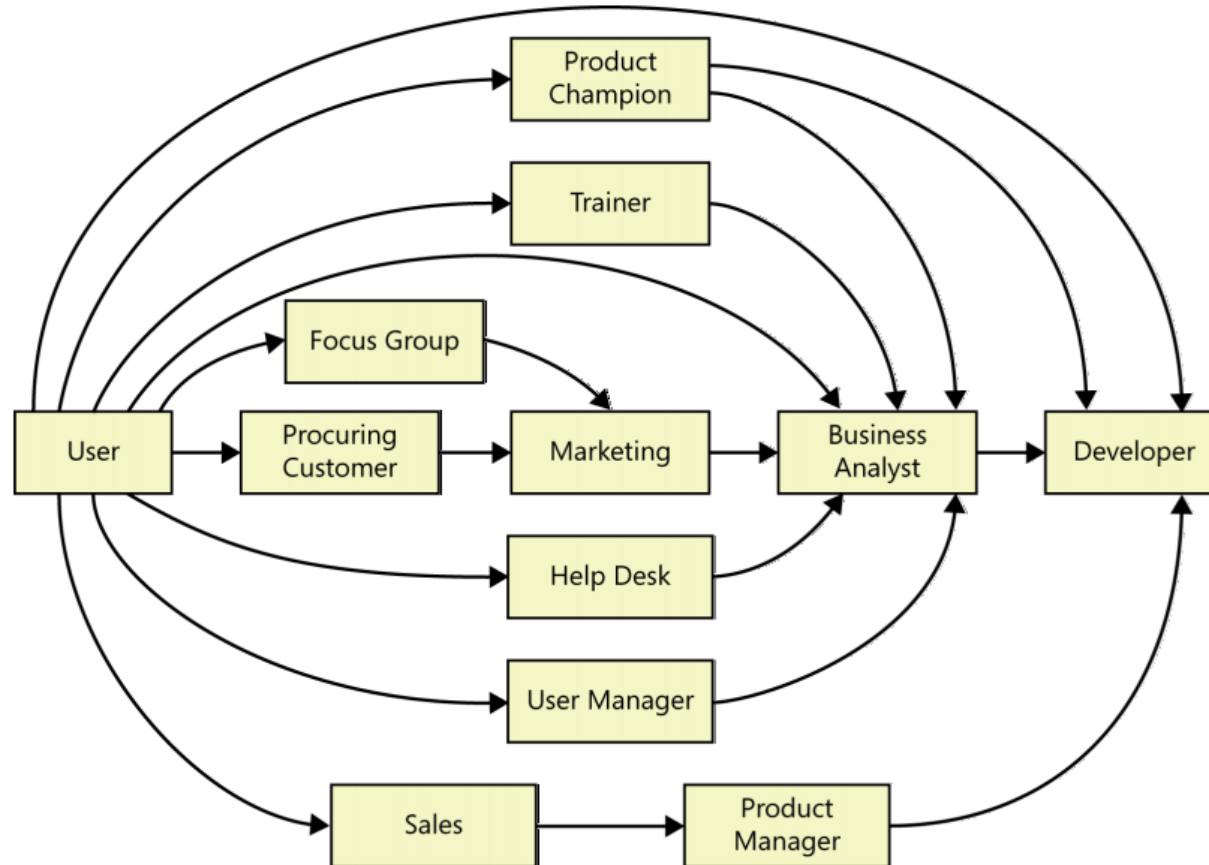
- A persona is a description of a hypothetical, generic person who serves as a stand-in for a group of users having similar characteristics and needs.
- Using personas understand the requirements and to design the user experience to best meet the needs of specific user communities.
- Working with a persona makes the requirements thought process more tangible than if you simply contemplate what a whole faceless group of people might want. Some people choose a random human face of the appropriate gender to make a persona seem even more real.

Connecting with user representatives

- Why do each user class needs someone to speak for requirements?
- Typical communication pathways between users and developers

Connecting with user representatives

- Typical communication pathways between users and developers



Product champion

- Definition: product champions?
 - Product Champions are few key members of our user community to provide the requirement
 - Each product champion serves as the primary interface between members of a single user class and the project's BA
- When developing commercial software, it can be difficult to find product champions from outside your company
 - Internal subject matter experts or outside consultants
 - Hire a suitable product champion who has the right background

Product champion

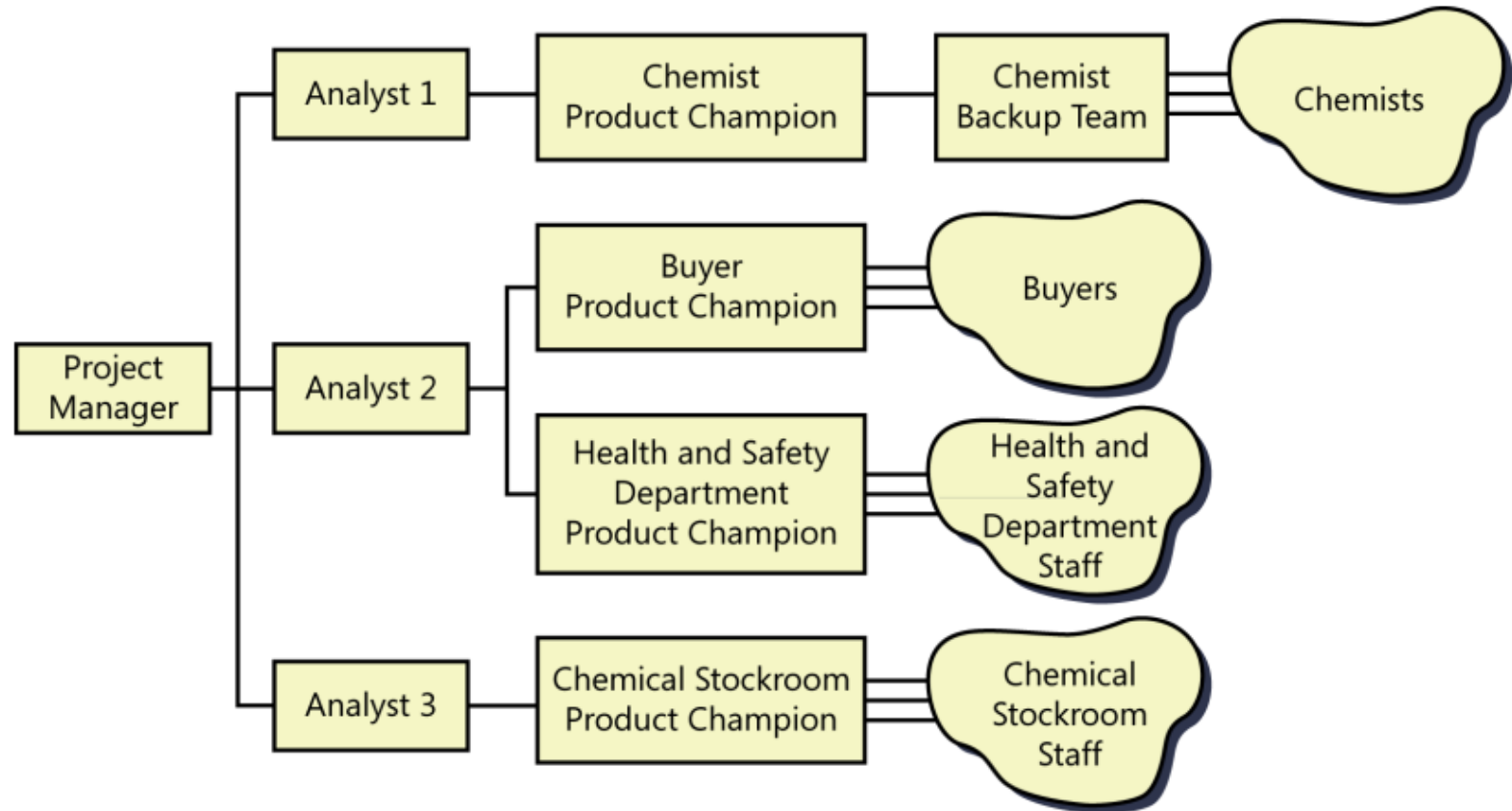
Product champion expectations

Category	Activities
Planning	■ Refine the scope and limitations of the product. ■ Identify other systems with which to interact. ■ Evaluate the impact of the new system on business operations. ■ Define a transition path from current applications or manual operations. ■ Identify relevant standards and certification requirements.
Requirements	■ Collect input on requirements from other users. ■ Develop usage scenarios, use cases, and user stories. ■ Resolve conflicts between proposed requirements within the user class. ■ Define implementation priorities. ■ Provide input regarding performance and other quality requirements. ■ Evaluate prototypes. ■ Work with other decision makers to resolve conflicts among requirements from different stakeholders. ■ Provide specialized algorithms.
Validation and verification	■ Review requirements specifications. ■ Define acceptance criteria. ■ Develop user acceptance tests from usage scenarios. ■ Provide test data sets from the business. ■ Perform beta testing or user acceptance testing.
User aids	■ Write portions of user documentation and help text. ■ Contribute to training materials or tutorials. ■ Demonstrate the system to peers.
Change management	■ Evaluate and prioritize defect corrections and enhancement requests. ■ Dynamically adjust the scope of future releases or iterations. ■ Evaluate the impact of proposed changes on users and business processes. ■ Participate in making change decisions.

Product champion

Multiple Product Champions

One person can rarely describe the needs for all users of an application. We need multiple product champions for major user classes



User representation on agile projects

- Product owner
- Frequent conversations
- Product owner vs. product champion
- An on-site customer doesn't guarantee the desired outcome

Resolving conflicting requirements

Disagreement between	How to resolve
Individual users	Product champion or product owner decides
User classes	Favored user class gets preference
Market segments	Segment with greatest impact on business success gets preference
Corporate customers	Business objectives dictate direction
Users and user managers	Product owner or product champion for the user class decides
Development and customers	Customers get preference, but in alignment with business objectives
Development and marketing	Marketing gets preference

